

Mathematics Stage 3 -- Year B -- Unit 34 Measurement — Mass, Length, Perimeter & Decimals

KLA: Mathematics | Stage 3 | Year 6 | Term: _____ | Duration: 2-3 weeks | Teacher: _____ | Year: _____

Syllabus outcomes

Code	Outcome descriptor
MAO-WM-01	Develops understanding and fluency through exploring, applying and communicating mathematical reasoning.
MA3-NSM-01	Selects and uses the appropriate unit and device to measure the masses of objects.
MA3-GM-02	Selects and uses the appropriate unit and device to measure lengths and distances including perimeters.
MA3-RN-02	Compares and orders decimals up to 3 decimal places.
MA3-RQF-02	Determines $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{5}$ and $\frac{1}{3}$ of measures and quantities; connects fractions, decimals and percentages.
MA3-AR-01	Selects and applies appropriate strategies to solve addition and subtraction problems (including with decimals).
MA4-ARE / LEN	Stage 4 (group A): area of triangles, parallelograms and composite shapes (ARE); perimeter of composite shapes and circumference of circles (LEN).

Syllabus content points (addressed through SOLO tracker)

All content points below are covered by students working independently through the SOLO tracker. Mini masterclasses target specific outcomes based on live tracker data.

Red -- Foundation	Yellow -- On-level	Green -- Extension
Measurement & number recall (single procedures) Recognise % means per cent; swap between benchmark fractions, decimals and percentages. Convert between kilograms and grams, and between kilograms and tonnes. Read and record mass and length using decimal notation to 3 decimal places (whole-number/decimal equivalence). Generate the whole quantity from a non-unit fractional part.	Measurement (applying, multi-step & word problems) Solve problems involving different units of mass. Convert between metres and kilometres and compare lengths/distances in the same unit. Investigate and compare perimeters of rectangles with the same area. Determine the number of different rectangles for a given area. Solve length and perimeter problems, including problems with different units of length. Solve word problems involving the addition and subtraction of decimals; justify the strategy.	Extended -- Stage 4 Area & Length Use formulas to find the area of triangles and parallelograms (MA4-ARE-C-01). Find the area of composite shapes and convert between area units (cm^2 , m^2) (MA4-ARE-C-01). Find the perimeter of composite shapes and the circumference of circles using $C = \pi d$ (MA4-LEN-C-01).

Teaching model

Students work independently through the Word Labs SOLO Tracker (wordlabs.app/solo) across differentiated bands -- Grow (resources), Know (practice), Show (auto-marked assessments). The teacher monitors live dashboard data and pulls small groups for targeted mini masterclasses (15-40 min) based on which outcomes have the most incomplete Show results. Remaining session time is used for incidental one-on-one and small group teaching in response to student questions and tracker data.

Working Mathematically (MAO-WM-01)

Process	How it appears in this unit
Communicating	Students describe units of mass and length, justify which unit is most suitable, and explain conversion and perimeter strategies using clear mathematical language.
Understanding and fluency	Students convert between metric units, record measurements in decimal notation to 3 dp, and add and subtract decimals accurately.
Reasoning	Students reason about the relationship between the size of a unit and the number of units, and about rectangles with equal area but different perimeter.
Problem solving	Students solve multi-step mass, length and perimeter word problems, and (Stage 4) area and circumference problems.

Differentiation

Support	HPGE / Extension
Students in this group:	Students in this group:

Pre-test

Date	
Format	
Band place ments / key finding s	

Post-test / unit evaluation

Date	
Key finding s	
Students who need further consolidation	
Next steps	

Resources

See Appendix for Resources

Word Labs SOLO Tracker: wordlabs.app/solo | NSW DoE Stage 3 Year B Unit 34 | NESA Mathematics K-10 Syllabus (2022)

Unit 34 -- Outcome Teaching Record

Find the outcome/s you taught, write your notes, sign and date. Order follows the outcome sequence, not your teaching calendar.

Co de	Outcome	Syllabus content point/s addressed	Mini masterclass teaching notes	Sign	Date
R1	Recognise % means per cent; connect benchmark fractions, decimals and percentages.	RQF/RN: recognise that % means percent; identify and use equivalent percentages, decimals and fractions to solve problems MA3-RQF-02 / MA3-RN-02 Year B	Mini Lesson 1		
R2	Convert between metric units of mass (kg ↔ g, kg ↔ tonnes).	NSM-B Mass: convert between kilograms and grams and between kilograms and tonnes MA3-NSM-01 Year B	Mini Lesson 2		
R3	Read and record mass and length in decimal notation to 3 decimal places.	Measurement decimals (mass = NSM-A, length = GM-B): recognise whole-number/decimal equivalence of measurements; interpret and record mass and length using decimal notation up to 3 dp MA3-RN-02 / MA3-RN-01 Year A & B	Mini Lesson 4		
R4	Generate the whole quantity from a non-unit fractional part.	RQF: build up to the whole — generate the whole quantity from non-unit fractional parts MA3-RQF-02 Year B	Mini Lesson 6		
Y1	Solve problems involving different units of mass.	NSM-B Mass: solve problems involving different units of mass MA3-NSM-01 Year B	Mini Lesson 3		
Y2	Convert between metres and kilometres and compare lengths in the same unit.	GM-B Length: use the decimal place value system to convert between metres and kilometres; convert measurements to the same unit to compare lengths and distances; relate the size of a unit to the number of units MA3-GM-02 Year B	Mini Lesson 10		
Y3	Investigate and compare perimeters of rectangles with the same area.	GM-B Length/perimeter: investigate and compare perimeters of rectangles with the same area MA3-GM-02 Year B	Mini Lesson 7		
Y4	Determine the number of different rectangles for a given area.	GM-B Length/perimeter: determine the number of different rectangles that can be formed using whole-number dimensions for a given area MA3-GM-02 Year B	Mini Lesson 8		
Y5	Solve length and perimeter problems, including different units of length.	GM-B Length/perimeter: solve a variety of problems involving length and perimeter, including problems involving different units of length MA3-GM-02 Year B	Mini Lesson 9		
Y6	Solve word problems adding and subtracting decimals; justify the strategy.	AR-B: solve word problems involving the addition and subtraction of decimals; justify why the strategy used is appropriate MA3-AR-01 Year B	Mini Lesson 5		
G1	Use formulas to find the area of triangles and parallelograms.	Stage 4 ARE-A: establish and use the formula to find the area of a parallelogram and of a triangle MA4-ARE-C-01 Stage 4 (group A)	Mini Lesson 11		
G2	Find the area of composite shapes; convert between area units (cm ² , m ²).	Stage 4 ARE-A: find the area of composite figures by dissection; convert between metric units of area MA4-ARE-C-01 Stage 4 (group A)	Mini Lesson 12		

Co de	Outcome	Syllabus content point/s addressed	Mini masterclass teaching notes	Sign	Date
G3	Find the perimeter of composite shapes and the circumference of circles.	Stage 4 LEN-A: find the perimeter of composite plane shapes; calculate the circumference of a circle using $C = \pi d$ MA4-LEN-C-01 Stage 4 (group A)	Mini Lesson 13		

Syllabus outcomes and content from Mathematics K-10 Syllabus (2022) copyright NSW Education Standards Authority (NESA) for and on behalf of the Crown in right of the State of New South Wales, 2024. NSM = Non-spatial measure (mass). GM = Geometric measure (length/perimeter). RN = Represents numbers (decimals). RQF = Representing quantity fractions. AR = Additive relations. ARE/LEN = Area / Length (Stage 4). A/B = syllabus content groups (B is the new measurement content for this Year-B unit).

Unit 34 -- Mini Lesson Sequence

Measurement — Mass, Length, Perimeter & Decimals | Stage 3 Year B

Supplement to the SOLO Teacher Program (pages 1-2). Print pages 1-2 only for annotation. Reference these lesson plans when planning mini masterclasses or when a reviewer requires evidence of planned lesson content.

Red band -- Foundation -- Mini Lessons 1-6

Yellow band -- On-level -- Mini Lessons 3-10

Green band -- Extension -- Mini Lessons 11-13

DoE lesson links: Lessons marked DoE Lesson X draw directly from NSW DoE Mathematics Stage 3 Unit 34. Steps have been condensed to 20-40 minutes for mini masterclass delivery. Resources referenced by their original resource numbers from the DoE unit.

Mini Lesson 1 Fractions, decimals and percentages

R1 | Red band | DoE Lessons 1-3 (adapted)

Duration: 20-25 min

Outcomes: R1

Syllabus: MA3-RQF-02, MA3-RN-02, MAO-WM-01 | FDP: % means per cent; equivalent fractions, decimals, percentages

Linked to: DoE Lessons 1-3 (adapted) -- Lessons 1-3 -- Daily number sense: fractions, decimals and percentages | Key resources: Resource 1 - grid paper, individual whiteboards

Learning intention

Students are learning to:
make connections between benchmark fractions, decimals and percentages

Success criteria

Students can:

- * I know % means 'per cent' (out of 100).
- * I can match benchmark fractions, decimals and percentages (e.g. $\frac{1}{2} = 50\% = 0.5$).
- * I can use equivalent forms to solve a simple problem.

Mini masterclass structure (20-25 min)

Activate (3 min): Shade $\frac{1}{2}$ of a 10x10 grid; ask the percentage and decimal. Establish % = out of 100.

Model (10 min): Benchmark table: $\frac{1}{2} = 50\% = 0.5$; $\frac{1}{4} = 25\% = 0.25$; $\frac{3}{4} = 75\% = 0.75$.

Guided practice (8 min): Students convert between benchmark fractions, decimals and percentages (e.g. 50% of 60).

Check (3 min): Exit: write 0.75 as a fraction and a percentage.

Key vocabulary

per cent, %, fraction, decimal, percentage, equivalent, benchmark, out of 100

Differentiation

Support: Use a shaded 10x10 grid; start with $\frac{1}{2}$ and $\frac{1}{4}$.
Extension: Convert 12 out of 60 to a fraction, decimal and percentage.

Assessment

Students connect benchmark fractions, decimals and percentages — links to R1.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 2 Converting units of mass [HANDS-ON]

R2 | Red band | DoE Lesson 1 (adapted)

Duration: 25-30 min

Outcomes: R2

Syllabus: MA3-NSM-01, MAO-WM-01 | NSM-B: convert between kilograms and grams and between kilograms and tonnes

Linked to: DoE Lesson 1 (adapted) -- Lesson 1 -- Pumpkin mass | Key resources: Equal-arm balances or digital scales, metric weights, Resource 2 - world's largest pumpkins

Physical materials needed: equal-arm balances or digital scales, metric weights

Learning intention

Students are learning to:
convert between common metric units of mass

Success criteria

Students can:

- * I know 1 kg = 1000 g and 1 tonne = 1000 kg.
- * I can convert between grams and kilograms.
- * I can convert between kilograms and tonnes.

Mini masterclass structure (25-30 min)

Activate (4 min): Heft and weigh a pumpkin piece. Record its mass in grams, then ask how to write the same mass in kilograms.

Model (10 min): Convert 2500 g = 2.5 kg ($\div 1000$), 3 kg = 3000 g ($\times 1000$), 1200 kg = 1.2 t ($\div 1000$). Use a place-value chart to show the $\times 1000$ / $\div 1000$ shift.

Guided practice (10 min): Students convert masses between g, kg and tonnes using scales and the world's-largest-pumpkins data.

Check (4 min): Exit question: 4500 g = ____ kg, and 2.3 t = ____ kg.

Key vocabulary

mass, gram, kilogram, tonne, convert, $\times 1000$, divide by 1000, place value

Differentiation

Support: Provide a g-kg-t conversion chart; use scales to read masses in both g and kg.
Extension: Convert masses given in decimal tonnes (e.g. 0.085 t) to kilograms and grams.

Assessment

Students convert between metric units of mass — links to R2 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 3 Mass problems with different units

Y1 | Yellow band | DoE Lessons 2 & 3 (adapted)

Duration: 30 min | **Outcomes: Y1** | Syllabus: MA3-NSM-01, MAO-WM-01 | NSM-B: solve problems involving different units of mass

Linked to: DoE Lessons 2 & 3 (adapted) -- Lessons 2 & 3 -- Which unit is best? / mass word problems | Key resources: Resource 4 - grams, kilograms, tonnes?, Resource 6 - household waste, individual whiteboards

Learning intention		Success criteria	
Students are learning to: solve problems involving different units of mass		Students can: * I can choose the most suitable unit (g, kg or tonnes) for an object. * I can convert so masses are in the same unit before comparing or adding. * I can solve a word problem that mixes units of mass.	
Mini masterclass structure (30 min)			
Activate (4 min): Sort items (apple, school bag, car, dining table) to the best unit of mass. Discuss why a dining table is best in kg, not g. Model (10 min): Solve a mixed-unit problem: a bin holds 8 kg, another holds 3500 g — total in kg? Convert first (3.5 kg), then add (11.5 kg). Guided practice (12 min): Students solve household-waste problems, converting between kg and tonnes and comparing masses across units. Check (4 min): Exit question: a recycling centre collects 4.5 t of paper in bags of 250 kg — how many bags?			
Key vocabulary		Differentiation	
mass, unit, gram, kilogram, tonne, convert, compare, total, difference		Support: Provide the converted values so students focus on choosing the unit and solving. Extension: Solve multi-step problems mixing all three units (g, kg, t) with decimals.	
Assessment		Teaching notes (annotate here)	
Students solve problems involving different units of mass — links to Y1 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:	

Mini Lesson 4 Decimal notation for measurements

R3 | Red band | DoE Lessons 3 & 4 (adapted)

Duration: 25-30 min

Outcomes: R3

Syllabus: MA3-RN-02, MA3-RN-01, MAO-WM-01 | RN-B: decimal notation for masses and lengths up to 3 dp

Linked to: DoE Lessons 3 & 4 (adapted) -- Lessons 3 & 4 -- Crocodile facts and measurements | Key resources: Place-value chart, Resource 11 - crocodile measurements, Resource 12 - crocodile conversion chart

Learning intention

Students are learning to:
connect decimal representations to the metric system and record measurements to 3 decimal places

Success criteria

Students can:

- * I know a whole-number and a decimal can name the same measurement (e.g. 2500 g = 2.5 kg).
- * I can interpret a mass or length written in decimal notation.
- * I can record a mass or length using decimal notation up to 3 decimal places.

Mini masterclass structure (25-30 min)

Activate (4 min): Show 1 m 250 mm on a place-value chart. Ask how to write it as a decimal number of metres (1.250 m).

Model (10 min): Record 2 kg 350 g = 2.350 kg, 4 m 75 cm = 4.75 m, 1 m 8 mm = 1.008 m. Stress each place value (tenths, hundredths, thousandths).

Guided practice (10 min): Students convert crocodile masses and lengths into decimal notation to 3 dp and back.

Check (4 min): Exit question: write 3 kg 60 g in kilograms as a decimal, and 2.045 m in metres and millimetres.

Key vocabulary

decimal notation, tenths, hundredths, thousandths, place value, equivalent, mass, length

Differentiation

Support: Use a labelled place-value chart with columns for tenths/hundredths/thousandths.
Extension: Order a set of measurements written in mixed units by first writing each to 3 dp.

Assessment

Students record measurements in decimal notation to 3 dp — links to R3 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 5 Adding and subtracting decimals

Y6 | Yellow band | DoE Lessons 2, 3 & 6 (adapted)

Duration: 30 min | **Outcomes:** Y6 | Syllabus: MA3-AR-01, MAO-WM-01 | AR-B: add and subtract decimals to solve word problems; justify the strategy

Linked to: DoE Lessons 2, 3 & 6 (adapted) -- Lessons 2, 3 & 6 -- Decimal addition and subtraction strategies | Key resources: Decimal place-value chart, individual whiteboards

Learning intention		Success criteria	
Students are learning to: solve word problems involving the addition and subtraction of decimals and justify the strategy used		Students can: * I can line up decimal points to add and subtract decimals. * I can choose an efficient strategy (levelling, constant difference, bridging). * I can justify why my strategy works for the problem.	
Mini masterclass structure (30 min)			
Activate (4 min): Show $2.65 + 1.4$. Ask why lining up the decimal points (and the place values) matters.			
Model (10 min): Add $12.75 + 8.6$ and subtract $9.3 - 4.85$ using a written strategy; then use constant difference for $6.02 - 5.97$. Justify each choice.			
Guided practice (12 min): Students solve decimal word problems (lengths, masses, money), choosing and justifying a strategy.			
Check (4 min): Exit question: a ribbon is 3.4 m; 1.85 m is cut off — how much is left? Show your strategy.			
Key vocabulary		Differentiation	
decimal, decimal point, place value, levelling, constant difference, bridging, justify, strategy		Support: Use a place-value chart and grid so digits line up; start with one-decimal-place numbers. Extension: Solve multi-step decimal problems mixing addition and subtraction, and explain the most efficient order.	
Assessment		Teaching notes (annotate here)	
Students add and subtract decimals in word problems and justify the strategy — links to Y6 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:	

Mini Lesson 6 Building the whole from a part

R4 | Red band | DoE Lessons 5-7 (adapted)

Duration: 20-25 min

Outcomes: R4

Syllabus: MA3-RQF-02, MAO-WM-01 | RQF: generate the whole quantity from non-unit fractional parts

Linked to: DoE Lessons 5-7 (adapted) -- Lessons 5-7 -- Daily number sense: build up to the whole | Key resources: Fraction bars, counters

Learning intention

Students are learning to:
generate the whole quantity from a non-unit fractional part

Success criteria

Students can:

- * I can find one part when I know a non-unit fraction (e.g. if $\frac{3}{4}$ is 18, one quarter is 6).
- * I can build up to the whole from the part.
- * I can check by finding the fraction of my answer.

Mini masterclass structure (20-25 min)

Activate (3 min): Tell students $\frac{3}{4}$ of a bag of marbles is 18. Ask how many marbles are in the whole bag.

Model (9 min): Use a fraction bar: $\frac{3}{4} = 18$, so each quarter ($\div 3$) = 6, and the whole ($\times 4$) = 24. Repeat with $\frac{2}{5} = 10 \rightarrow$ whole = 25.

Guided practice (10 min): Students find the whole from a given non-unit fractional part using a bar model, then check.

Check (3 min): Exit question: $\frac{2}{3}$ of a length is 12 m. How long is the whole length?

Key vocabulary

fraction, non-unit fraction, part, whole, unit fraction, build up, divide, multiply

Differentiation

Support: Draw the fraction bar and find one part first; use counters for the part.

Extension: Find the whole from a part for fractions like $\frac{3}{5}$ and $\frac{5}{8}$, then find a different fraction of that whole.

Assessment

Students generate the whole from a fractional part — links to R4 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 7 Same area, different perimeter [HANDS-ON]

Y3 | Yellow band | DoE Lessons 5 & 6 (adapted)

Duration: 30-35 min **Outcomes:** Y3 Syllabus: MA3-GM-02, MAO-WM-01 | GM-B: investigate and compare perimeters of rectangles with the same area

Linked to: DoE Lessons 5 & 6 (adapted) -- Lessons 5 & 6 -- Different shapes, same perimeter / Darryl's fencing | Key resources: 1 cm grid paper, 30 cm rulers, individual whiteboards

Physical materials needed: 1 cm grid paper, 30 cm rulers

Learning intention

Students are learning to:
investigate and compare perimeters of rectangles with the same area

Success criteria

Students can:

- * I can draw different rectangles that all have the same area.
- * I can find the perimeter of each rectangle.
- * I can compare the perimeters and say which is largest or smallest.

Mini masterclass structure (30-35 min)

Activate (4 min): On grid paper, draw a rectangle with area 12 cm². Ask whether another rectangle of area 12 cm² has the same perimeter.

Model (12 min): Draw all rectangles of area 12 cm² (1x12, 2x6, 3x4) and find each perimeter (26, 16, 14 cm). Notice the closer to a square, the smaller the perimeter.

Guided practice (14 min): Students investigate rectangles with a fixed area (e.g. 24 cm²), record perimeters in a table, and compare.

Check (4 min): Exit question: two rectangles both have area 16 cm². Could they have different perimeters? Give an example.

Key vocabulary

area, perimeter, rectangle, dimensions, length, width, compare, square

Differentiation

Support: Provide grid paper with the area outlined; help count perimeter by ones around the edge.
Extension: For a fixed area, predict which rectangle gives the smallest perimeter and explain why.

Assessment

Students compare perimeters of rectangles with the same area — links to Y3 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 8 Rectangles for a given area **[HANDS-ON]**

Y4 | Yellow band | DoE Lessons 5 & 6 (adapted)

Duration: 30 min **Outcomes:** Y4 Syllabus: MA3-GM-02, MAO-WM-01 | GM-B: determine the number of different rectangles for a given area

Linked to: DoE Lessons 5 & 6 (adapted) -- Lessons 5 & 6 -- Different shapes, same perimeter | Key resources: 1 cm grid paper, individual whiteboards

Physical materials needed: 1 cm grid paper

Learning intention		Success criteria	
Students are learning to: determine the number of different rectangles that can be formed using whole-number dimensions for a given area		Students can: * I can use factors to find whole-number side lengths for a given area. * I can list all the different rectangles with that area. * I can say how many different rectangles there are.	
Mini masterclass structure (30 min)			
Activate (4 min): Ask how many different whole-number rectangles have an area of 12 cm ² . Connect to factor pairs of 12. Model (10 min): List factor pairs of 12 (1x12, 2x6, 3x4) -> 3 rectangles. Repeat for area 16 (1x16, 2x8, 4x4) -> 3 rectangles. Treat 2x6 and 6x2 as the same. Guided practice (12 min): Students find all rectangles for given areas (e.g. 18, 24, 36 cm ²) using factor pairs and count them. Check (4 min): Exit question: how many different rectangles (whole-number sides) have an area of 20 cm ² ? List them.			
Key vocabulary		Differentiation	
area, rectangle, factor, factor pair, dimensions, whole number, square		Support: Provide a factor list for the target number; build each rectangle on grid paper. Extension: Find an area with exactly 2 (or exactly 4) different rectangles and explain using factors.	
Assessment		Teaching notes (annotate here)	
Students determine the number of rectangles for a given area — links to Y4 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:	

Mini Lesson 9 Length and perimeter problems

Y5 | Yellow band | DoE Lessons 5-8 (adapted)

Duration: 30-35 min **Outcomes:** Y5 Syllabus: MA3-GM-02, MAO-WM-01 | GM-B: solve length and perimeter problems including different units

Linked to: DoE Lessons 5-8 (adapted) -- Lessons 5-8 -- Fencing, mystery bones and mega trees | Key resources: Resource 14 - missing perimeter problems, soft measuring tape, 30 cm rulers

Learning intention		Success criteria
Students are learning to: solve a variety of problems involving length and perimeter, including problems involving different units of length		Students can: * I can find a missing side length using the perimeter. * I can solve perimeter problems where lengths are in different units. * I can solve a real-life length or perimeter problem.
Mini masterclass structure (30-35 min)		
Activate (4 min): A rectangle has perimeter 30 cm and one side 7 cm. Ask how to find the other side. Model (12 min): Find a missing side from the perimeter; then a fencing problem mixing units (sides 1.2 m and 80 cm) — convert to the same unit first, then add. Guided practice (14 min): Students solve missing-side, mixed-unit and real-life perimeter problems (gardens, fences), converting units as needed. Check (4 min): Exit question: a garden has a perimeter of 15.5 m with 4 sides — give one possible set of side lengths.		
Key vocabulary		Differentiation
length, perimeter, side, convert, unit, metre, centimetre, missing side, total		Support: Provide the conversion and a labelled diagram; find one missing side at a time. Extension: Design an irregular shape with a given perimeter and justify the side lengths chosen.
Assessment		Teaching notes (annotate here)
Students solve length and perimeter problems with different units — links to Y5 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:

Mini Lesson 10 Converting and comparing lengths

Y2 | Yellow band | DoE Lessons 7 & 8 (adapted)

Duration: 30 min | **Outcomes:** Y2 | Syllabus: MA3-GM-02, MAO-WM-01 | GM-B: convert between metres and kilometres; compare lengths in the same unit

Linked to: DoE Lessons 7 & 8 (adapted) -- Lessons 7 & 8 -- Mystery bones and mega trees | Key resources: Resource 20 - conversion chart, trundle wheels, soft measuring tape

Learning intention

Students are learning to:
use the decimal place value system to convert between metres and kilometres and compare lengths and distances

Success criteria

Students can:

- * I know 1 km = 1000 m and can convert between metres and kilometres.
- * I can convert measurements to the same unit before comparing them.
- * I understand that a larger unit needs a smaller measure for the same length.

Mini masterclass structure (30 min)

Activate (4 min): Show 2500 m. Ask how to write it in kilometres (2.5 km). Discuss why the number gets smaller when the unit gets bigger.

Model (10 min): Convert 1.8 km = 1800 m, 3450 m = 3.45 km. Compare distances by converting to the same unit (e.g. 0.9 km vs 950 m).

Guided practice (12 min): Students convert m<->km using the conversion chart and order a set of distances given in mixed units.

Check (4 min): Exit question: order 1.2 km, 950 m and 1100 m from shortest to longest.

Key vocabulary

length, distance, metre, kilometre, convert, place value, compare, order, unit

Differentiation

Support: Provide a m-km conversion chart; convert everything to metres before comparing.
Extension: Convert tree heights and diameters to mm and cm, and investigate relationships between the measurements.

Assessment

Students convert and compare metric units of length — links to Y2 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 11 Area of triangles and parallelograms (Stage 4)

G1 | Green band | Original mini lesson

Duration: 30-35 min

Outcomes: G1

Syllabus: MA4-ARE-C-01, MAO-WM-01 | Stage 4 ARE-A: area of triangles and parallelograms

Learning intention

Students are learning to:
use formulas to find the area of triangles and parallelograms

Success criteria

Students can:

- * I can find the area of a parallelogram using base x height.
- * I can find the area of a triangle using $\frac{1}{2} \times \text{base} \times \text{height}$.
- * I can identify the perpendicular height to use.

Mini masterclass structure (30-35 min)

Activate (4 min): Show a parallelogram. Cut and rearrange it into a rectangle to see that area = base x height.

Model (12 min): Find the area of a parallelogram (base 8 cm, height 5 cm = 40 cm²) and a triangle ($\frac{1}{2} \times 8 \times 5 = 20$ cm²). Stress using the perpendicular height, not the slant side.

Guided practice (12 min): Students find areas of triangles and parallelograms from given base and height measurements.

Check (5 min): Exit question: find the area of a triangle with base 10 cm and height 6 cm.

Key vocabulary

area, base, perpendicular height, parallelogram, triangle, formula, square units

Differentiation

Support: Provide the formula and a labelled diagram; identify the height before substituting.
Extension: Find a missing base or height when given the area, by working backwards from the formula.

Assessment

Students find the area of triangles and parallelograms — links to G1 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 12 Composite area and area units (Stage 4)

G2 | Green band | Original mini lesson

Duration: 30-35 min

Outcomes: G2

Syllabus: MA4-ARE-C-01, MAO-WM-01 | Stage 4 ARE-A: composite area by dissection; convert area units

Learning intention

Students are learning to:
find the area of composite shapes and convert between area units

Success criteria

Students can:

- * I can split a composite (L-shape) figure into rectangles.
- * I can add the part areas to find the total area.
- * I can convert between cm^2 and m^2 .

Mini masterclass structure (30-35 min)

Activate (4 min): Show an L-shaped figure. Ask how we could split it to find its area.

Model (12 min): Dissect an L-shape into two rectangles, find each area, and add. Then convert: $1 \text{ m}^2 = 10\,000 \text{ cm}^2$, so $30\,000 \text{ cm}^2 = 3 \text{ m}^2$.

Guided practice (12 min): Students find areas of composite figures by dissection and convert areas between cm^2 and m^2 .

Check (5 min): Exit question: an L-shape splits into a 6×4 and a 3×2 rectangle — what is its total area?

Key vocabulary

composite figure, dissect, rectangle, area, total, square centimetre, square metre, convert

Differentiation

Support: Mark the dividing line on the figure; provide each rectangle's dimensions.
Extension: Find a missing length in a composite figure when the total area is given.

Assessment

Students find composite areas and convert area units — links to G2 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 13 Composite perimeter and circumference (Stage 4)		
G3 Green band Original mini lesson		
Duration: 30 min	Outcomes: G3	Syllabus: MA4-LEN-C-01, MAO-WM-01 Stage 4 LEN-A: perimeter of composite shapes; circumference of circles
Learning intention		Success criteria
Students are learning to: find the perimeter of composite shapes and the circumference of circles		Students can: * I can find the perimeter of a composite shape by finding all side lengths. * I know the circumference of a circle is $C = \pi \times \text{diameter}$. * I can calculate a circumference using $\pi \approx 3.14$.
Mini masterclass structure (30 min)		
Activate (4 min): Show an L-shaped figure with some sides missing. Ask how to work out the missing sides before adding the perimeter. Model (11 min): Find the perimeter of a composite shape by deducing missing sides. Then introduce $C = \pi d$: a circle with diameter 10 cm has $C \approx 3.14 \times 10 = 31.4$ cm. Guided practice (11 min): Students find perimeters of composite shapes and circumferences of circles (given the diameter) using $\pi \approx 3.14$. Check (4 min): Exit question: find the circumference of a circle with diameter 8 cm (use $\pi \approx 3.14$).		
Key vocabulary		Differentiation
perimeter, composite shape, missing side, circle, circumference, diameter, radius, pi (π)		Support: Provide the missing side lengths and the formula $C = \pi d$; use $\pi \approx 3.14$. Extension: Find the circumference from the radius ($C = 2\pi r$), and the perimeter of a shape made from a rectangle and a half-circle.
Assessment		Teaching notes (annotate here)
Students find composite perimeters and circumferences — links to G3 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:

Unit 34 -- Resource Appendix

Every link verified. Videos open in a browser; worksheets are print-ready PDFs (see the download checklist at the end).

Type	Resource	Link
R1 -- Recognise % means per cent; connect benchmark fractions, decimals and percentages.		
Video	Converting Between Fractions, Decimals, and Percents A Mini Course Math with Mr. J	https://www.youtube.com/watch?v=2vjbNZaFz3c
Video	Fractions to Percents Math with Mr. J	https://www.youtube.com/watch?v=XXf4wHb2lYc
Worksheet (print)	Corbettmaths — Fractions, Decimals, Percentages (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/fractions-decimals-percentages-pdf.pdf
R2 -- Convert between metric units of mass (kg ↔ g, kg ↔ tonnes).		
Video	Kilograms and Grams Converting kg to g and g to kg Math with Mr. J	https://www.youtube.com/watch?v=zXUICZRxDwQ
Video	Metric Units of Mass Convert mg, g, and kg Math with Mr. J	https://www.youtube.com/watch?v=ptrKThvQwh4
Worksheet (print)	Corbettmaths — Metric Units (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/metric-units-pdf.pdf
R3 -- Read and record mass and length in decimal notation to 3 decimal places.		
Video	Place Value to the Thousandths — Representing Decimal Numbers Miacademy	https://www.youtube.com/watch?v=BsZJGMRWEK0
Video	Expanded Form with Decimals Math with Mr. J	https://www.youtube.com/watch?v=GX8o-S6Vxig
Worksheet (print)	Corbettmaths — Place Value (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/place-value-pdf.pdf
Worksheet (print)	Corbettmaths — Ordering Decimals (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/ordering-decimals-pdf1.pdf
R4 -- Generate the whole quantity from a non-unit fractional part.		
Video	Fractions: Finding the Original Corbettmaths	https://www.youtube.com/watch?v=pdUg0UwXBHk
Video	Fractions of Numbers: Finding the Whole Partners in Prime	https://www.youtube.com/watch?v=x0ljSSMfZXw
Worksheet (print)	Corbettmaths — Fractions of Amounts (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/fractions-of-amounts-pdf.pdf
Y1 -- Solve problems involving different units of mass.		
Video	Metric Mass Measurements — Grams, Kilograms, and Tonnes Miacademy	https://www.youtube.com/watch?v=Nek4OfX8zKE
Video	Metric Units of Mass Convert mg, g, and kg Math with Mr. J	https://www.youtube.com/watch?v=ptrKThvQwh4
Worksheet (print)	Corbettmaths — Metric Units (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/metric-units-pdf.pdf
Y2 -- Convert between metres and kilometres and compare lengths in the same unit.		
Video	Metric Units of Length Convert mm, cm, m and km Math with Mr. J	https://www.youtube.com/watch?v=kOJFSH_Bn9U
Video	Meters and Centimeters Converting m to cm and cm to m Math with Mr. J	https://www.youtube.com/watch?v=gALZRL4v2NM
Worksheet (print)	Corbettmaths — Metric Units (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/metric-units-pdf.pdf
Y3 -- Investigate and compare perimeters of rectangles with the same area.		
Video	Same Area, Different Perimeter Miss Hutchison	https://www.youtube.com/watch?v=bsl-r7dRSnc
Video	How to Find the Perimeter of a Rectangle Math with Mr. J	https://www.youtube.com/watch?v=teMDmMxy550
Worksheet (print)	Corbettmaths — Perimeter (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/perimeter-pdf.pdf
Y4 -- Determine the number of different rectangles for a given area.		
Video	Finding the Area of a Rectangle Using the Formula Math with Mr. J	https://www.youtube.com/watch?v=1Ugk12yTQM
Video	How to Find the Area of Rectangles and Squares Math with Mr. J	https://www.youtube.com/watch?v=_ejDO6aPPzU
Worksheet (print)	Corbettmaths — Area of a Rectangle (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/area-of-a-rectangle-pdf.pdf

Type	Resource	Link
Y5 -- Solve length and perimeter problems, including different units of length.		
Video	How to Find Perimeter Math with Mr. J	https://www.youtube.com/watch?v=TSLc5p2VuYM
Video	Finding the Perimeter and Area of a Composite Shape Math with Mr. J	https://www.youtube.com/watch?v=rBpsauKXPfs
Worksheet (print)	Corbettmaths — Perimeter (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/perimeter-pdf.pdf
Y6 -- Solve word problems adding and subtracting decimals; justify the strategy.		
Video	Adding and Subtracting Decimals (How to) Math with Mr. J	https://www.youtube.com/watch?v=PnwLv6khwk8
Video	How to Add and Subtract Decimals (Step-by-Step Examples) Math with Mr. J	https://www.youtube.com/watch?v=Cg- TeiaSa8
Worksheet (print)	Corbettmaths — Adding Decimals (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/adding-decimals-pdf.pdf
Worksheet (print)	Corbettmaths — Subtracting Decimals (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/subtracting-decimals-pdf.pdf
G1 -- Use formulas to find the area of triangles and parallelograms.		
Video	Finding the Area of a Triangle A Step-By-Step Guide Math with Mr. J	https://www.youtube.com/watch?v=-ZPFB1uDjEw
Video	Area of Parallelograms How to Find the Area of a Parallelogram Math with Mr. J	https://www.youtube.com/watch?v=uj6k22WubCk
Worksheet (print)	Corbettmaths — Area of a Triangle (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/area-of-a-triangle-pdf.pdf
Worksheet (print)	Corbettmaths — Area of a Parallelogram (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/area-of-a-parallelogram-pdf.pdf
G2 -- Find the area of composite shapes; convert between area units (cm², m²).		
Video	How to Find the Area of a Composite Shape L-Shaped Example Math with Mr. J	https://www.youtube.com/watch?v=LAXHgQwcrDc
Video	Finding the Area of Composite Figures with Triangles Math with Mr. J	https://www.youtube.com/watch?v=bo483Varm-U
Worksheet (print)	Corbettmaths — Area of Compound Shapes (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/area-of-compound-shapes-pdf.pdf
G3 -- Find the perimeter of composite shapes and the circumference of circles.		
Video	Perimeter of Composite Figures Math with Mr. J	https://www.youtube.com/watch?v=4c_-Q5gADcQ
Video	Circumference of a Circle Math with Mr. J	https://www.youtube.com/watch?v=_E0C5ECDS0U
Worksheet (print)	Corbettmaths — Circumference (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/circumference-pdf.pdf
Beyond SOLO -- Stage 4 extension		
Video	Finding the Area of a Triangle Math with Mr. J	https://www.youtube.com/watch?v=-ZPFB1uDjEw
Video	Area of Parallelograms Math with Mr. J	https://www.youtube.com/watch?v=uj6k22WubCk
Video	How to Find the Area of a Composite Shape Math with Mr. J	https://www.youtube.com/watch?v=LAXHgQwcrDc
Video	Circumference of a Circle Math with Mr. J	https://www.youtube.com/watch?v=_E0C5ECDS0U
Worksheet (print)	Corbettmaths — Area of a Triangle (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/area-of-a-triangle-pdf.pdf
Worksheet (print)	Corbettmaths — Area of Compound Shapes (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/area-of-compound-shapes-pdf.pdf
Worksheet (print)	Corbettmaths — Circumference (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/circumference-pdf.pdf
Website	Maths is Fun — Area (interactive)	https://www.mathsisfun.com/area.html
Website	Maths is Fun — Circle (interactive)	https://www.mathsisfun.com/geometry/circle.html

Worksheet download checklist

Download and print these PDFs before the unit (or save to your class drive). Tick each as you go.

- ☐ Corbettmaths — Fractions, Decimals, Percentages (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/fractions-decimals-percentages-pdf.pdf>
- ☐ Corbettmaths — Metric Units (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/metric-units-pdf.pdf>
- ☐ Corbettmaths — Place Value (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/place-value-pdf.pdf>

- ☐ Corbettmaths — Ordering Decimals (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/ordering-decimals-pdf1.pdf>
- ☐ Corbettmaths — Fractions of Amounts (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/fractions-of-amounts-pdf.pdf>
- ☐ Corbettmaths — Perimeter (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/perimeter-pdf.pdf>
- ☐ Corbettmaths — Area of a Rectangle (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/area-of-a-rectangle-pdf.pdf>
- ☐ Corbettmaths — Adding Decimals (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/adding-decimals-pdf.pdf>
- ☐ Corbettmaths — Subtracting Decimals (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/subtracting-decimals-pdf.pdf>
- ☐ Corbettmaths — Area of a Triangle (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/area-of-a-triangle-pdf.pdf>
- ☐ Corbettmaths — Area of a Parallelogram (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/area-of-a-parallelogram-pdf.pdf>
- ☐ Corbettmaths — Area of Compound Shapes (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/area-of-compound-shapes-pdf.pdf>
- ☐ Corbettmaths — Circumference (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/circumference-pdf.pdf>